
Guidelines for writing reports

A report has to be written on every experiment. This report should be written in the report book provided.

Pay attention to grammatical correctness and do not mix languages. Marks will be awarded for neatness and logical presentation of your data.

i. Heading

Use the three-letter code and description of the specific experiment as the heading.

ii. Objective

Briefly state the objectives of the experiment.

iii. Results

It is not necessary to describe the theory, experimental set up and method! Only your observations, results and conclusions should be reported in a systematic and organized way.

Give all your results and wherever possible, make use of tables and graphs, **even when not specifically requested to do so**. All graphs must be drawn according to the hints in your practical manual - see experiment 2.

Always take the **accuracy** with which your **measurements** were taken into account and round your answers off to the **correct number of significant figures!**

iv. Discussion and Conclusions

A brief discussion of your observations and results, even when not requested, is required. A few points to consider are:

- Are your results in agreement with the theory?
 - What are the reasons for any observed deviation from the ideal result or behaviour?
 - Can you think of experimental errors which could have influenced your results?
 - Is it possible to estimate the effect of experimental errors on your observations and results?
 - Do you have suggestions on how the results could have been improved? Do you have any suggestions or own theories?
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Units

Every measurement of a physical quantity must be expressed in the applicable SI unit. The SI system is based on the units given in the table below. It then follows that although measurements in an experiment should be given in the units of the instrument, it is necessary to convert these units to **SI units** before they are used in calculations.

S.I. Units

The international system of units (Système International d'Unités) is based on the mksa (meter-kilogram-second-ampere) system. The fundamental quantities are:

Quantity	Variable	Unit symbol	Unit
<i>length</i>	ℓ	m	metre
<i>mass</i>	m	kg	kilogram
<i>time</i>	t	s	second
<i>electric current</i>	I	A	ampere
<i>temperature</i>	T	K	kelvin
<i>luminous intensity</i>	L	cd	candela
<i>amount of substance</i>	n	mol	mole

All other quantities, can be defined in terms of the fundamental quantities.

Quantity	Variable	Unit Symbol	Unit
<i>area</i>	A	m ²	square metre
<i>volume</i>	V	m ³	cubic metre
<i>frequency</i>	F	Hz	hertz
<i>density</i>	ρ	kg/m ³	kilogram per cubic metre
<i>speed, velocity</i>	v	m/s	metre per second
<i>angular velocity</i>	ω	rad/s	radian per second
<i>acceleration</i>	a	m/s ²	metre per second squared
<i>angular acceleration</i>	α	rad/s ²	radian per second squared
<i>force</i>	F	N	newton
<i>pressure</i>	p	Pa	pascal
<i>work, energy, quantity of heat</i>	W	J	joule
<i>power</i>	P	W	watt
<i>electric charge</i>	Q	C	coulomb
<i>potential difference, electromotive force</i>	V	V	volt
<i>voltage</i>	V	V	volt
<i>electrical resistance</i>	R	Ω	ohm
<i>capacitance</i>	C	F	farad
<i>inductance</i>	L	H	henry

Multiple and sub-units

<i>Factor</i>	10^9	10^6	10^3	10^{-3}	10^{-6}	10^{-9}
<i>Prefix</i>	giga	mega	kilo	milli	micro	nano
<i>Symbol</i>	G	M	k	m	μ	n

Errors

Every measurement contains an amount of uncertainty in it, that is, no measurement is 100% accurate. This is referred to as the error in a measurement. In the first place, the measuring instrument itself sets a limit on the how accurate a reading can be. A ruler for example, that is subdivided into centimeters, will measure the length of an object correctly only to the nearest centimeter. Its error will therefore be 0.5 cm (otherwise the measurement would have been estimated to the other centimeter mark). This type of error is called the **instrumental error**. Apart from instrumental errors we also have observational and personal errors that come into play - especially the errors of parallax and interpolation estimations. The determination of the error of a measurement is so important that one should supply both the measurement and an estimation of the error of that measurement, for example the diameter of a cylinder was measured as

$$d = 4.40 \pm 0.05 \text{ cm}$$

The 0.05 cm is called the **absolute error**. We could also write

$$d = 4.40 (1 \pm 0.05/4.40) \text{ cm}$$

where $0.05/4.40$ is called the **relative error** of the measurement.

The **percentage error** is expressed as a percentage of the actual value:

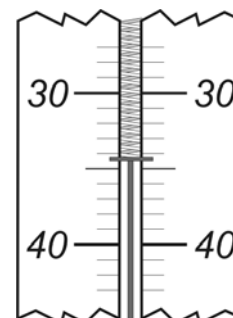
$$\%error = \frac{\text{measured value} - \text{actual value}}{\text{actual value}} \times 100\%$$

There are methods to determine the error of a result. It is however, a difficult subject and for our purposes it is adequate to round the result off to the appropriate amount of **significant figures**.

Significant figures

Reading an analogue scale

When taking a reading from an analogue scale you should estimate the least significant digit of the measurement by mentally subdividing the smallest division on the scale into more intervals. The diagram depicts part of a spring scale from which an object is suspended. The scale indicates the mass of the object in grams. One would report the measurement as **34.4 gram** in this case. Note that the final digit of the measurement (4/10 of a gram) was estimated and has an (not too unrealistic) amount of uncertainty. The suspended object does have a definite mass which we refer to as the **true mass** of the object. The measuring device attempts to determine this value. The **accuracy** of such a measurement is the amount by which the measurement differs from the true value. If the pointer of the spring scale came to rest exactly on the 34 gram mark on the scale one should report the mass as 34.0 gram. In such an instance the final digit would still have a certain measure of uncertainty. This reflects the situation that a measurement is only an approximation of the true value being measured. For analogue scales the measurement therefore should have one more digit than that given by the smallest division on the scale. Note that readings from a graph also entail reading an analogue scale.

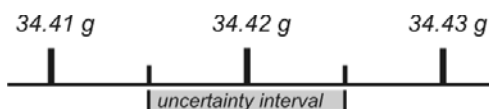


Reading a digital scale

Suppose that we weigh the same object on an electronic scale with a **resolution** of 0.01 gram and the scale indicates its mass as in the diagram. Even in this case there is still a measure of uncertainty in the measurement since the **true mass**, m , of the object can be anywhere within the range

34.42 gram

$$34.415 < m < 34.425$$



and the scale would still report the mass as 34.42 grams. The true mass of the object can be anywhere in the indicated interval according to the measurement. That is the mass can be reported as 34.420 ± 0.005 grams. In short it should be reported as 34.420 gram. So in this case we also add an additional significant digit to the measurement albeit a zero in all cases for a digital scale. The uncertainty will again be in the least significant digit of the measurement. This final zero is therefore one digit beyond the resolution of the instrument.

When are digits significant?

- All non-zero digits are significant, for example 27 N has two significant figures and 3.46 N has three significant figures.

When zeros appear in measurements one needs to consider their position in the value in order to decide whether they are significant.

- Leading zeros are not significant, they just serve to locate the decimal point. By using scientific notation one can get rid of such zeros. For example 0.034 gram has only two significant figures.
- Zeros sandwiched between other digits are significant, so 20.03 °C has four significant figures.
- Zeros at the end of a measurement are considered not to be significant, that is 4 500 metres has only two significant figures. If the experimenter needs to indicate that they are actually significant then he/she should use scientific notation - see the next remark.
- Zeros after the decimal point are considered to be significant. That is 12.0 A has three significant figures and 4.500×10^3 m has four significant figures.

Significant figures and measurements

In the example above we see that measuring the object with the spring scale gave us a mass of 34.4 grams while the electronic scale gave us a mass of 34.420 grams. The price of the spring scale is about R50 and that of the electronic scale R5000. So the extra two digits in the second measurement came at a substantial price. In the experimental world it usually takes much more than just money to get a more accurate value for a certain measurement. It usually involves more time, effort and ingenuity to get those extra digits. **Take care in reporting measurements to the correct number of significant figures.**

Calculations with measurements

Let's assume that we are actually interested in the weight, W , of the object. It can be calculated using the value 9.8 m/s^2 for gravitational acceleration. Since the second measurement came at a higher price one would expect that the weight calculated from its measurement should have more significant figures to it. That is for the spring scale we would expect that the weight should be reported as 0.337 N and for the electronic scale 0.33732 N, while keeping the answers to the same number of significant figures as the measurements. In actuality the gravitational acceleration is also a measurement with its own uncertainty. It follows logically that the uncertainty in its value will also have an influence on the number of digits in the calculated weight of the object. So how does one decide how many significant figures there should be in a value calculated using measurements?

Significant figures in calculated values

We use some simple rules to decide how many significant figures the final value should be rounded off to. In **calculations involving multiplication, division, trigonometric and other functions** we round the final value off to the same number of significant figures as the measurement with the least number of significant figures. The weight of the object would therefore be reported as 0.34 N in both cases due to the fact that the gravitational acceleration has only two significant figures. The situation can be improved upon if we can obtain a value for g with more significant figures (more money, time and effort). As a further example if we were to calculate the value of $\sin(kx)$ using the values $k = 0.097 \text{ m}^{-1}$ and $x = 4.73 \text{ m}$ we obtain the answer 0.44 since k has only two significant figures.

When the value being calculated is obtained from **additions and/or subtractions** of values the final value is rounded off to the same **decimal** as the measurement with the least number of decimals, for example:

5.67 J	2 decimal places
1.1 J	1 decimal place
0.9378 J	4 decimal places
7.7 J	1 decimal place

Exact values

Numerical constants have no uncertainty in their value. They are therefore not measurements and do not influence the uncertainty in a value calculated from them. Some examples are the number of millimeters in an inch (2.54 mm), which is a defined quantity (the inch) or π , which can be calculated to any required precision or the number 2 (a counting value) in the formula $2\pi r$. Such values are not taken into consideration when deciding how many significant figures there should be in the final answer.

Intermediate calculations

When a calculated value is the result of several steps of calculations one should keep (at least) one more significant digit in the intermediate answers than that which is required in the final value.

Scientific notation

Very large or small values are best reported using scientific notation. In this notation the decimal point is repositioned just after the first significant digit in the value. For example to report 14297 J in scientific notation the decimal point needs to be positioned just after the digit 1. That is we need to divide the value by ten thousand or multiply it by 10^{-4} . In order not to change the value we therefore also need to multiply it by ten thousand or 10^4 :

$$14297 J$$

$$= 14297 \times 10^{-4} \times 10^{+4} J$$

$$= 1.4297 \times 10^4 J$$

Graph Rules

Rules for drawing graphs

1. The graph should have a concise title. All symbols that are used in the graph must be mentioned in words in the heading and the subject or apparatus of the experiment should be mentioned, e.g. The temperature change, ΔT , of a sample of water as heat, Q_w , is added to it.
2. The independent variable (the quantity that is changed **directly** by the experimenter) is plotted on the horizontal axis (the abscissa) and the dependent variable (the quantity that changes in the experiment as a result of changing the independent variable) is plotted on the vertical axis (the ordinate).
3. Clearly indicate along each axis the variable plotted on the axis in words as well as the relevant units e.g. resistance(Ω).
4. The scales of the two axes, not necessarily the same, should be chosen so as to allow the graph to occupy most of the available space. The axes do not need to start at zero.
5. The intervals should be chosen to **facilitate** plotting and reading of points.
6. Mark each experimental point with a dot with a circle around it.
7. If the points lie more or less on a straight line use a ruler to draw a straight line in such a position that the points are evenly spaced about it. This is called the line of best fit. Otherwise draw a smooth curve through the points. This reduces the effect of experimental errors. Never connect the points by means of short straight lines.

8. The gradient of a graph

The gradient (or slope) of a graph is **always** calculated by means of the definition:

$$\text{gradient} = \frac{y_2 - y_1}{x_2 - x_1}$$

This definition is also used when the graph passes through the origin. **Never** calculate the slope using $m = y/x$.

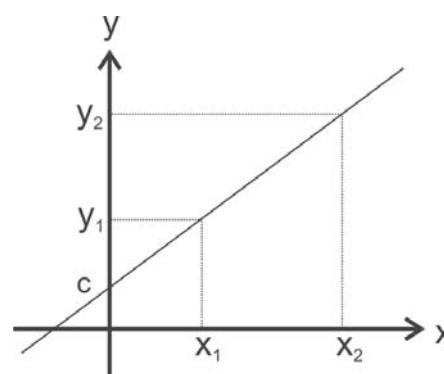


Fig.1 a Linear relationship.

- Always determine the slope from the fitted line, **never from the experimental points**.
- Choose the largest possible Δx and Δy , this allows for more accurate results. In other words, choose two points that are as far apart as possible.
- If the slope of any region of a graph is not constant, it is called a curve, as in **figure 2**. The slope of a curve at any given point is equal to the slope of the tangent at that point. The slope of the tangent is calculated in the same manner as for a straight line. The gradient of a curve is therefore not constant and depends on the independent variable.

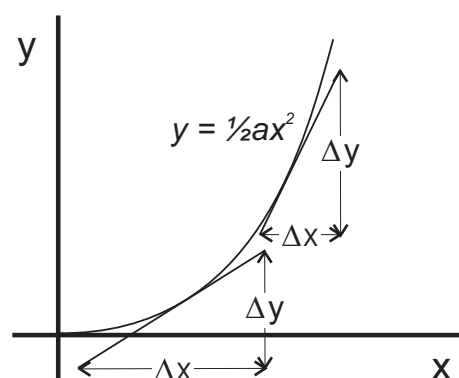
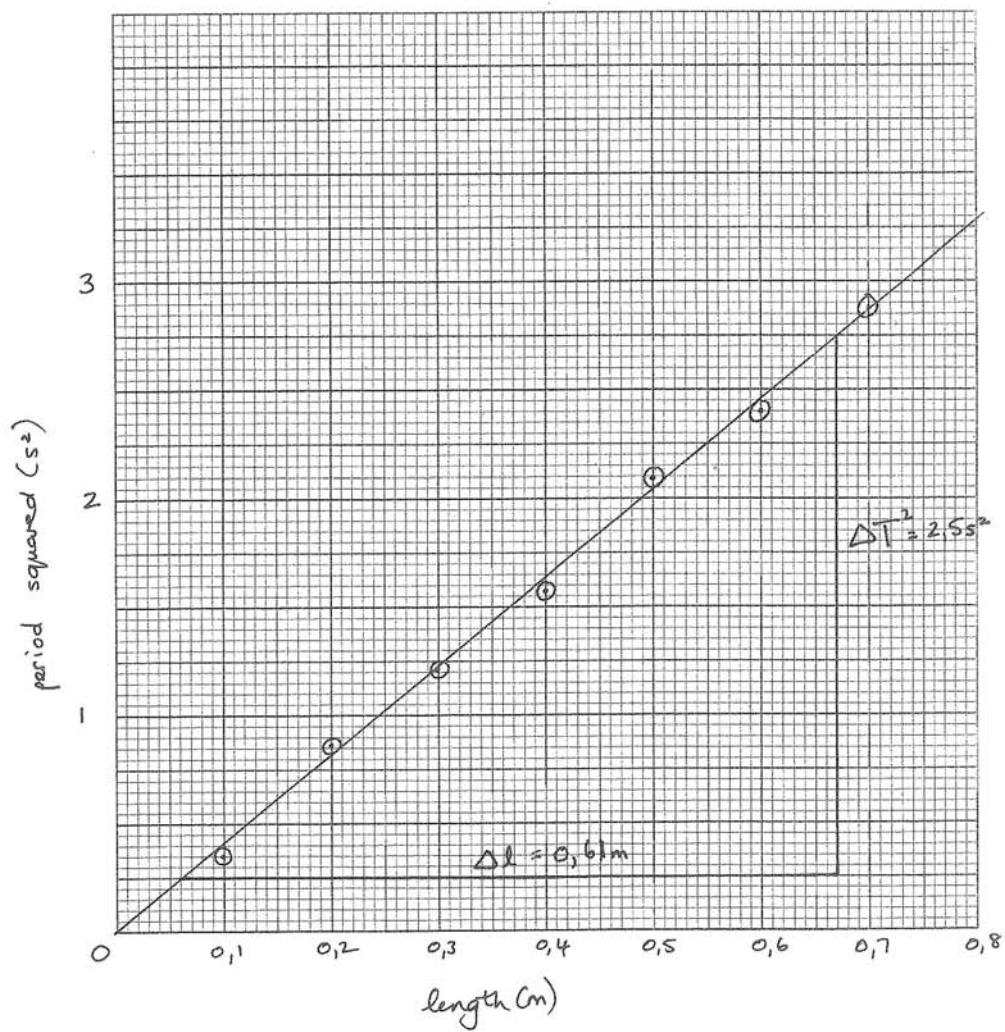


Fig. 2 The gradient of a curve.

9. If the relationship between the dependent and the independent variable is linear as in **figure 1**, then the relationship between the two quantities is of the form:

$$y = mx + c$$

The square of the period of a simple pendulum as a function of its length.



where m is the gradient of the graph and c is the y-intercept. This equation with the aforementioned constants is called the empirical (obtained through experiment procedures) equation. By comparing the empirical equation with its theoretical counterpart it is possible to determine some useful constants.

The empirical equation of the graph in the example on page 10 is

$$T^2 = 4,1(s^2/m)\ell + 0$$

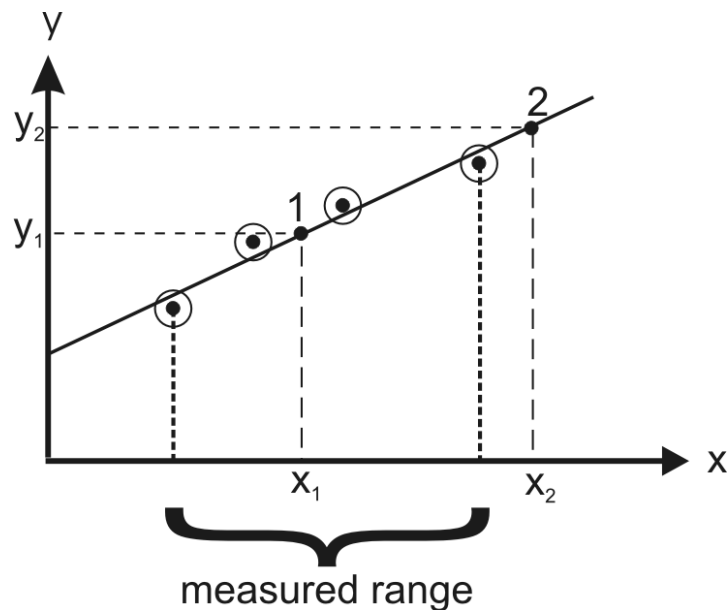


Fig.3 Interpolation and extrapolation.

10. A graph can be used to read off values for the dependent variable at values of the independent variable at which no measurement were made, like at x_1 and x_2 in **figure 3**.

If the reading is made at a value for the independent variable that is within the experimental measured range then the process is called **interpolation**, as at point 1. Interpolation leads to relatively reliable values for the dependent variable.

If the reading is made at an independent value outside of the measured range then the process is called **extrapolation**, as at point 2. Such values are not as reliable as those obtained by interpolation since the functional relationship may deviate from a linear relationship outside the measured range. Rather extend the measuring range to include the value at which a reading is required if this is at all possible.

References:

1. Fishbane: Physics for Scientists and Engineers
2. Giancoli: Physics
3. Halliday: Fundamentals of Physics
4. Young et al: University Physics
5. Halliday et al: Fundamentals of Physics

